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Latch assembly for use with an inversion drum

Abstract

A hinged cover overlays an opening that provides access to a pressurized inversion drum. A releasable latch assembly holds the cover in a closed and locked position. The latch assembly comprises a plurality of first knuckles formed in the cover that are interlaceable with a plurality of second knuckles formed in the body adjacent the opening. The latch assembly further comprises a rotatable latch pin that may be installed within the first and second knuckles. The latch assembly is moved from an unlocked to a locked position by rotation of the latch pin within the knuckles.

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Exhibit “A” is a photograph of an inversion drum with a latch assembly that was known in the art and offered for sale in the United States more than 1 year prior to Aug. 23, 2017. The inversion drum shown in Exhibit “A” is manufactured by IBC HydroTech. More information can be found at <https://www.ibg-hydro-tech.de/en/products/lining/inversion-drums/>. cited by applicant

Exhibit “B” is a photograph of an inversion drum with a latch assembly that was known in the art and offered for sale in the United States more than 1 year prior to Aug. 23, 2017. The inversion

drum shown in Exhibit "B" is manufactured by Hammerhead. More information can be found at <https://www.hammerheadtrenchless.com/products/lateral-cipp/inversion-drums>. cited by applicant Exhibit "C" is a photograph of an inversion drum with a latch assembly that was known in the art and offered for sale in the United States more than 1 year prior to Aug. 23, 2017. The inversion drum shown in Exhibit "C" is manufactured by Fluvius. More information can be found at <http://www.fluvius.info/en/waste-water/full-length-repair/inversion-units.html>. cited by applicant

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Background/Summary

SUMMARY

(1) The present invention is directed to an apparatus comprising a body having an opening formed therein, a cover sized to cover the opening, and a latch assembly. The latch assembly comprises a plurality of spaced and aligned first knuckles formed in a selected one of the cover and body, each first knuckle having an exposed semi-cylindrical groove formed therein, and a plurality of spaced and aligned second knuckles formed in the unselected one of the cover and body adjacent the opening. The latch assembly further comprises a rotatable latch pin having aligned and alternating first and second sections. Each first section of the latch pin has a size and shape complementary to the semi-cylindrical groove of a first knuckle.

(2) The present invention is also directed to an apparatus comprising a body having an opening formed therein, a cover sized to cover the opening, and a latch assembly. The latch assembly comprises a plurality of knuckles formed in a selected one of the cover and body and a plurality of knuckles formed in the unselected one of the cover and body adjacent the opening. The plurality of first knuckles are configured to interlace with the plurality of second knuckles. The latch assembly further comprises a rotatable latch pin having a plurality of half-moon cross-sections.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an inversion drum utilizing a latch assembly of the present invention. The latch assembly is shown in an unlocked position.

(2) FIG. 2 is a perspective view of a body and cover configured for use with the latch assembly. The cover is shown in an open position and the latch assembly is shown in an unlocked position. A latch pin used with the latch assembly has been removed for clarity.

(3) FIG. 3 is a top plan view of the body and cover shown in FIG. 2.

(4) FIG. 4 is a cross-sectional view of a portion of the latch assembly formed in the body, taken along line A-A from FIG. 3.

(5) FIG. 5 is a perspective view of the latch pin used with the latch assembly.

(6) FIG. 6 is a perspective view of the body and cover shown in FIG. 3. The cover is shown in the open position and the latch assembly is shown in an unlocked position. The latch pin is shown installed within the body.

(7) FIG. 7 is a top plan view of the body and cover shown in FIG. 6.

(8) FIG. 8 is a cross-sectional view of a portion of the latch assembly formed in the body, taken

along line B-B from FIG. 7.

(9) FIG. 9 is a top plan view of the cover in a closed position. The latch assembly is shown in an engaged position.

(10) FIG. 10 is a cross-sectional view of the latch assembly in the engaged position, taken along line C-C from FIG. 9.

(11) FIG. 11 is an enlarged view of area D shown in FIG. 10.

(12) FIG. 12 is a top plan view of the cover in the closed position. The latch assembly is shown in a locked position.

(13) FIG. 13 is a cross-sectional view of the latch assembly in the locked position, taken along line E-E from FIG. 12.

(14) FIG. 14 is an enlarged view of area F shown in FIG. 13.

(15) FIG. 15 is a perspective view of an inversion drum utilizing an alternative embodiment of the latch assembly. The latch assembly is shown in a locked position.

(16) FIG. 16 is an enlarged view of area G shown in FIG. 15.

(17) FIG. 17 is a perspective view of the inversion drum shown in FIG. 15. A portion of the drum has been cut-away for better display. The latch assembly is shown in an unlocked position.

(18) FIG. 18 is an enlarged view of area H shown in FIG. 17.

(19) FIG. 19 is an enlarged view of area I shown in FIG. 18.

DETAILED DESCRIPTION

(20) Cured in Place Piping (CIPP) operations are performed to repair existing underground pipes. The operation involves the installation of a resin saturated, tubular liner within the existing pipe. The resin soaked liner cures against the interior of the pipe and seals any cracks or damaged areas.

(21) A pressurized inversion drum is used to install the liner within the existing pipe. The interior of the liner is typically made of felt while the outside of the liner is typically made of a water resistant material. In operation, the felt interior of the tubular liner is first filled with resin. The resin-filled liner is flattened and installed within the drum by coiling it around a spindle supported within the drum. Once the liner is installed in the drum, the interior of the drum may be pressurized.

(22) The pressure inside the drum operates to invert and expand the liner into the existing pipe as it is discharged from the drum. Because the liner is inverted as it expands the length of the existing pipe, the resin soaked surface of the liner contacts the interior walls of the pipe. Water may be pumped into the interior of the inverted liner to help expand the liner throughout the pipe as the resin cures adjacent to the pipe walls. In alternative embodiments, steam or UV light may be used to cure the resin to the pipe walls.

(23) The resin used to saturate the liner is typically mixed with a hardening agent. If the hardening agent sets before the liner is installed within the pipe, the liner is no longer usable. Thus, time is of the essence with performing CIPP operations.

(24) Inversion drums known in the art have access ports that are sealed using multiple threaded fasteners. Each fastener may have a knob that requires a dozen or so revolutions before it becomes unfastened. Such fasteners take time to engage or release. The covers used with such access ports also typically need to be completely detached from the drum prior to installing the liner within the drum.

(25) The present invention is directed to a latch assembly that allows rapid access to the interior of an inversion drum to quickly install a tubular liner within the drum. The latch assembly also allows for rapid sealing of the drum in order to quickly pressurize the interior of the drum.

(26) With reference to FIG. 1, a pressurized inversion drum 10 is shown. The inversion drum 10 comprises a cylindrical body 12 joined to a discharge cone 14. The body 12 is supported on a frame 16 and may be transported using a set of wheels 18 attached to the frame.

(27) A tubular liner (not shown) is installed in the drum 10 through an access port 20. Once inside the drum 10, the liner is wound around a spindle 22 by a handwheel 24. The liner is discharged from the drum 10 in an inverted form through the discharge cone 14. An air regulator 26, ball valve

28, and pressure gage **30** are supported on the top of the body **12**. These items allow an operator to pressurize the drum **10** for the inversion process. The operator may view the inside of the drum **10** using a first sight glass **32** installed in the drum body **12**.

(28) The access port **20** is supported on the top of the drum **10** and comprises a body **34** sealed by a cover **36**. The cover **36** may be opened and closed using a handle **38** attached to the cover. A second sight glass **40** is installed in the cover **36** in order to allow the operator to view the inside of the drum **10**. A set of hose connections **42**, **44** are also installed in the cover **36**. The hose connections **42**, **44** allow water to be circulated throughout the drum **10** to either retard or accelerate setup of the resin contained within the liner.

(29) Turning to FIG. 2, the body **34** of the access port **20** comprises a standoff **46** having a central opening **48** formed therein. As shown in FIG. 1, the bottom edge of the standoff **46** is attached to the body **12** of the drum **10** and surrounds an opening (not shown) formed in the drum body **12**. The tubular liner is installed within the drum **10** by passing through the opening **48** formed in the standoff **46** and the opening formed in the body **12** of the drum **10**.

(30) A flange **50** is supported on the top edge of the standoff **46** that surrounds the central opening **48**. The flange **50** is bounded by a first end **52**, an opposite second end **54**, and side edges **56**. Likewise, the cover **36** is bounded by a first end **58**, an opposite second end **60**, and side edges **62**. Both the cover **36** and flange **50** have knuckles **64**, **66** formed on their second ends **54**, **60** that are configured to mate and receive a pin **68**. Installation of the pin **68** within the knuckles **64**, **66** rotatably secures the cover **36** to the flange **50** in the form of a hinged joint **70**.

(31) In alternative embodiments, the standoff **46** may not be used. Instead the flange **50** may be formed as part of the body **12** of the drum **10**. In such embodiment, the flange **50** would surround the opening (not shown) formed in the drum body **12**.

(32) FIG. 2 shows the cover **36** in an open position. The cover **36** is shown in a closed position in FIG. 1. The cover **36** rotates between an open and closed position via the hinged joint **70**. When closed, the cover **36** is sized to cover the opening **48** and mate with the flange **50** such that the side edges **62** of the cover **36** are flush with the side edges **56** of the flange **50**.

(33) The cover **36** is locked to the body **34** by a latch assembly **72**, the entirety of which is shown in FIGS. 10 and 13. With reference to FIGS. 2-4, a portion of the latch assembly **72** comprises a plurality of spaced and aligned first knuckles **74** formed in the first end **58** of the cover **36**. Each of the first knuckles **74** extends downwards from the bottom surface of the first end **58** of the cover **36**. A first exposed semi-cylindrical groove **76** is formed in the inner surface of each of the first knuckles **74**. Each of the grooves **76** opens towards the second end **60** of the cover **36**. In one embodiment, the outer surface of each first knuckle **74** is rounded.

(34) The latch assembly **72**, further comprises a plurality of spaced and aligned second knuckles **78** formed in the first end **52** of the flange **50** adjacent the opening **48**. Each of the second knuckles **78** is integral with the first end **52** of the flange **50**. A short cylindrical passage **80** extends through each second knuckle **78**. In one embodiment, the outer surface of each second knuckle **78** is rounded.

(35) A notch **82** is formed in the first end **52** of the flange **50** between each second knuckle **78**. Each of the notches **82** are bounded on each side by adjacent second knuckles **78**. A second exposed semi-cylindrical groove **84** is formed in the flange **50** within each notch **82**. Each of the grooves **84** extends the length of each notch **82** and joins each passage **80** formed in each second knuckle **78**, as shown in FIG. 4.

(36) The plurality of first knuckles **74** are spaced so that they are vertically alignable with each of the notches **82** in a one-to-one relationship. Thus, when the cover **36** is rotated to a closed position, each of the first knuckles **74** is positioned within a corresponding one of the notches **82**. The plurality of first and second knuckles **74**, **78** are interlaced with one another when the cover **36** is in the closed position. In one embodiment, the outer surfaces of each first and second knuckle **74**, **78** are identical in shape and construction so as to form a smooth edge when interlaced.

(37) When the knuckles **74**, **78** are interlaced, each of the first semi-cylindrical grooves **76** aligns with a corresponding one of the second semi-cylindrical grooves **84** in a one-to-one relationship. The aligned grooves **76**, **84** join each of the short passages **80** so as to form a single passage, as shown in FIG. **13**. As shown in FIGS. **2-3**, each of the knuckles **74**, **78** may vary in length. In alternative embodiments, each of the knuckles **74**, **78** may be the same length.

(38) In alternative embodiments, the plurality of first knuckles **74** may be formed in the body **34** instead of the cover **36**. In such case, the plurality of second knuckles **78** are formed in the cover **36** instead of the body **34**.

(39) Turning to FIGS. **5-6**, the latch assembly **72** further comprises a rotatable latch pin **86**. The latch pin **86** has aligned and alternating first and second sections **88**, **90** that are bounded by a first and second end **92**, **94**. Each first section **88** has a semicircular cross-sectional shape resembling a half moon. Each first section **88** is complementary in size and shape to each of the first and second grooves **76**, **84**. The first sections **88** are aligned with the longitudinal axis of the latch pin **86**.

(40) Each of the second sections **90** of the latch pin **86** has a size and shape complementary to the short cylindrical passages **80** formed in each of the second knuckles **78**. The latch pin **86** is installed within the flange **50** so that it extends through each second knuckle **78**, as shown in FIG. **6**. The length of each second section **90** may correspond with the length of each second knuckle **78**.

(41) A handle **96** is rigidly attached to each opposed end **92**, **94** of the latch pin **86**, as shown in FIG. **6**. In one embodiment, each of the handles **96** is a rod that extends the length of each side edge **56** of the flange **50**. In alternative embodiments, only a single handle **96** may be attached to the latch pin **86**.

(42) Turning to FIGS. **7-8**, when the latch pin **86** is installed within the flange **50**, each first section **88** is positioned within each second groove **84** and each second section **90** is positioned within each short passage **80**. When the latch pin **86** is in this position, each of the notches **82** is clear to receive a corresponding one of the first knuckles **74**, as shown in FIG. **7**.

(43) The latch assembly **72** is in the unlocked position when the notches **82** are clear of the latch pin **86**. In this position, the handles **96** are positioned adjacent the side edges **56** of the flange **50**, as shown in FIG. **7**. When the notches **82** are clear of the latch pin **86**, the cover **36** may rotate between its open and closed positions, as shown in FIGS. **1-2**.

(44) Clockwise rotation of the handles **96** rotates the latch pin **86** clockwise within the flange **50**, and causes the latch assembly **72** to move from an unlocked to a locked position. Once the cover **36** is in a closed position, the first and second knuckles **74**, **78** are interlaced. With the cover **36** so positioned, the latch assembly **72** may be moved from the unlocked position to a locked position.

(45) During an intermediate stage of the transition between unlocked and locked positions, the latch assembly **72** reaches the engaged position shown in FIGS. **9-11**. Each of the first sections **88** has moved partially out of the second groove **84** and partially into the first groove **76**, as shown in FIGS. **10-11**. In this position, the latch pin **86** engages both the first and second knuckles **74**, **78**.

(46) Further clockwise rotation of the handles **96** moves the latch assembly **72** from the engaged position to the locked position shown in FIGS. **12-14**. Such rotation moves the first sections **88** entirely or substantially entirely within the first grooves **76**, as shown in FIGS. **13-14**. In the locked position of the cover **36**, the first knuckles are prevented from moving vertically within the notches **82**. Thus, the cover **36** is locked to the flange **50**.

(47) The latch assembly **72** may be moved to an unlocked position by counterclockwise rotation of the handles **96**, which causes each first section **88** to move out of a first groove **76** and into a second groove **84**. When the cover **36** is unlocked, the handles **96** are positioned adjacent the side edges **56** of the flange **50**.

(48) Once the first sections **88** are fully positioned in the first grooves **76**, the first knuckles **74** are free to move out of the notches **82**. Thus, the cover **36** is free to rotate about the hinge joint **70** to the open position, shown in FIGS. **2** and **6**.

(49) Rotation of the handles **96** through an included angle of 180 degrees moves the latch assembly

72 between its locked and unlocked positions. When the cover 36 is locked, as shown in FIG. 12, the handles extend away from both the cover 36 and the flange 50. When the cover 36 is unlocked, the handles are positioned adjacent the side edges 56 of the flange 50.

(50) When the interior of the drum 10 is under pressure, the locked latch assembly 72 can resist substantial opening forces applied to the cover 36. The resistance is attributable to the lack of eccentricity between the first sections 88 and the rotational axis of the latch pin 86. As a result, there is no tendency for the force applied to the interior of the cover 36 to rotate the latch pin 86 to the unlocked position.

(51) For example, if pressure within the drum is 40 psi, the latch assembly 72 may resist up to 5,000 pounds of force applied to the cover 36. Yet the latch assembly 72 needs only a relatively small amount of force to seal the cover 36 over the opening 48.

(52) With reference to FIGS. 6, 10-11, and 13-14, a seal 98 is positioned on the bottom surface of the cover 36. The seal 98 is sized to surround the opening 48, as shown in FIG. 6. The seal 98 engages the flange 50 when the cover 36 is in the closed position. In one embodiment, the seal 98 is an O-ring. Locking of the latch assembly 72 compresses the seal 98 tightly against the flange 50, so that pressure can be maintained in the drum 10.

(53) In alternative embodiments, the diameter of each first section 88 may be offset from the longitudinal axis of the latch pin 86. Such design provides a tighter lock between the first and second knuckles 74, 78 and in turn provides greater compression of the seal 96. Greater compression may be required if a larger seal is used in place of the O-ring shown herein.

(54) With reference to FIGS. 15-19, an alternative embodiment of the latch assembly 100 is shown for use with the inversion drum 10. The latch assembly 100 uses the same cover 36, body 34, and latch pin 86 used with latch assembly 72, shown in FIGS. 1-14. However, the latch assembly 100 utilizes a different handle system 102.

(55) Like handles 96, the handles 102 are rigidly attached to opposite ends 92, 94 of the latch pin 86. Each of the handles 102 is substantially rectangular in shape and has a series of grips 104 formed on its bottom surface. Unlike the handles 96, the handles 102 extend less than half the length of the side edges 56 of the flange 50.

(56) With reference to FIGS. 17-19, the latch assembly 100 operates the latch pin 86 in the same manner as latch assembly 72. However, the handles 102 are rotated clockwise to unlock the latch assembly 100, rather than to lock it. The first sections 88 are positioned within the second grooves 84 when the handles 102 are extended away from the cover 36 and flange 50.

(57) With reference to FIGS. 15-16, the latch assembly 100 is in the locked position when the handles 102 are adjacent the flange 50. Positioning the handles 102 adjacent the flange 50 when the latch assembly 100 is in the locked position allows each of the handles 102 to be secured to the flange 50 by an independent lock mechanism 105. Securing the handles 102 to the flange 50 provides a back-up lock in case the latch assembly 100 fails.

(58) In one embodiment, the lock mechanism 105 is a plunger 106 that may be disposed within a bore 108 formed in each of the handles 102, as shown in FIG. 18. The plunger 106 is then secured to an opening (not shown) formed in the side edge 52 of the flange 50. In alternative embodiments, the lock mechanism 105 may comprise a threaded fastener or clamp. In alternative embodiments, only one handle 102 may be used instead of two.

(59) Changes may be made in the construction, operation and arrangement of the various parts, elements, steps and procedures described herein without departing from the spirit and scope of the invention as described in the following claims.

Claims

1. An apparatus, comprising: a pressurized inversion drum comprising: a body having an opening formed therein, the opening surrounded by an outer body edge; a plurality of exposed notches

formed in the outer body edge; a cover having opposed inner and outer surfaces and an outer cover edge, the cover positioned over the opening and movable between an open position and a closed position; a rotatable latch pin; a rotatable handle coupled to the body; in which rotation of the handle moves the rotatable latch pin between an unlocked position and a locked position; a plurality of knuckles projecting from the outer cover edge; in which the outer cover edge is sealed against the outer body edge when the cover is in the closed position; in which the plurality of knuckles mate with the plurality of exposed notches in a one-to-one relationship when the cover is in the locked position thereby preventing the cover from being removed from the opening and in which the plurality of knuckles and plurality of exposed notches form a plurality of passages; in which the rotatable latch pin is installed within the plurality of passages; and in which the cover is configured to be removed from the opening when the latch pin is in the unlocked position; a groove formed within the inner surface of the cover; a seal installed within the groove; in which the seal engages the outer body edge when the cover is in the closed position; and a window formed within the cover such that an interior of the body is viewable through the window.

2. The apparatus of claim 1, in which the cover compresses the seal against the outer edge of the body when the latch pin is in the locked position.

3. The apparatus of claim 1, in which the plurality of knuckles are characterized as a first plurality of knuckles, and in which the body further comprises: a plurality of second knuckles joined to the outer body edge, each of the plurality of second knuckles having a passage formed therein; in which each of the plurality of exposed notches is positioned intermediate a pair of the plurality of second knuckles.

4. The apparatus of claim 3, in which rotation of the rotatable latch pin installs portions of the rotatable latch pin within the exposed notches.

5. The apparatus of claim 1, in which the plurality of knuckles are in a spaced-relationship.

6. The apparatus of claim 1, in which the plurality of exposed notches are in a spaced-relationship.

7. The apparatus of claim 1, in which the plurality of exposed notches are characterized as a first plurality of exposed notches; and a second exposed notch is formed within each of the plurality of knuckles, the second exposed notch configured to mate with a corresponding one of the first plurality of notches so as to form a cylindrical passage.

8. The apparatus of claim 1, in which the pressurized inversion drum further comprises: a rotatable spindle installed within the body and configured to wind an elongate liner within the body.

9. The apparatus of claim 1, in which the seal is compressed against the body in response to rotation of the rotatable handle.

10. The apparatus of claim 1 in which the plurality of exposed notches are clear of the plurality of knuckles when the cover is closed and the latch pin is in the unlocked position.

11. An apparatus, comprising: a pressurized inversion drum comprising: a body having an opening formed therein, the opening surrounded by an outer body edge; a plurality of exposed notches formed in the outer body edge; a cover having opposed inner and outer surfaces and an outer cover edge, the cover positioned over the opening and movable between an open position and a closed position; in which the outer cover edge is sealed against the outer body edge when the cover is in the closed position; a rotatable handle coupled to the body; in which rotation of the handle moves a latch pin from an unlocked position to a locked position; a plurality of knuckles projecting from the outer cover edge; in which the plurality of exposed notches are characterized as a first plurality of exposed notches, and a second exposed notch is formed within each of the plurality of knuckles, the second exposed notch configured to mate with a corresponding one of the first plurality of notches so as to form a passage; in which the plurality of knuckles are configured to mate with the plurality of exposed notches in a one-to-one relationship when the latch pin is in the locked position, thereby preventing the cover from being removed from the opening; and in which the cover is configured to be removed from the opening when the latch pin is in the unlocked position; a groove formed within the inner surface of the cover; a seal installed within the groove; in which

the cover compresses the seal against the outer body edge when the latch pin is in the locked position.

12. The apparatus of claim 11, in which the rotatable latch pin is installed within the passage.

13. The apparatus of claim 11, in which the plurality of knuckles are characterized as a first plurality of knuckles, and in which the body further comprises: a plurality of second knuckles joined to the outer body edge, each of the plurality of second knuckles having a passage formed therein; in which each of the plurality of exposed notches is positioned intermediate a pair of the plurality of second knuckles.

14. The apparatus of claim 13 in which: the rotatable latch pin is installed within the plurality of passages; in which rotation of the rotatable latch pin installs portions of the rotatable latch pin within the plurality of exposed notches.

15. The apparatus of claim 11, in which the plurality of knuckles are in a spaced-relationship.

16. The apparatus of claim 11, in which the plurality of exposed notches are in a spaced-relationship.

17. The apparatus of claim 11, in which the pressurized inversion drum further comprises: a window formed within the cover such that an interior of the body is viewable through the window.

18. The apparatus of claim 11, in which the pressurized inversion drum further comprises: a rotatable spindle installed within the body and configured to wind an elongate liner within the body.

19. The apparatus of claim 11, in which the seal is compressed against the body in response to rotation of the rotatable handle.

20. A method of using the apparatus of claim 11, comprising: placing the cover over the opening such that the plurality of knuckles are adjacent the plurality of exposed notches but not engaged; thereafter, rotating the rotatable handle such that the latch pin is moved from the unlocked position to the locked position.

21. The apparatus of claim 11 in which the plurality of exposed notches are adjacent the plurality of knuckles when the latch pin is in the unlocked position.

22. The apparatus of claim 1 in which the plurality of exposed notches are adjacent the plurality of knuckles when the latch pin is in the unlocked position.

23. The apparatus of claim 11 in which the plurality of exposed notches are clear of the plurality of knuckles when the cover is closed and the latch pin is in the unlocked position.
